

BETONIQ WEST LTD — INTERNAL FOUNDER DOCUMENT

The Offline Double-Spend Problem, Frame by Frame: What We Actually Solved, and What We Didn't

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0. Direct Answers to Your Two Questions

Lab validation: a separate, controlled-lab step — NOT a formal checkbox requirement inside the CBN Regulatory Sandbox application itself. It's the \$60,000 "Independent Lab Attack-Test Validation (FIME/UL-equivalent)" line already built into the Niger State \$500,000 ask. We're doing it proactively so we walk in with independent third-party proof, not just our own claim.

Have we cracked it: yes, and no — precisely stated below. We have NOT solved the general, provably-impossible version of offline double-spend that applies to ledger/account-based systems (nobody has, or ever will — a consensus-theory limit, not an engineering gap). We HAVE architected around the need to solve that problem at all, using an approach with 25+ years of real prior art behind its core idea, plus two genuinely new hardware security layers on top nobody has combined this way for rural offline NFC.

1. Why EMV / Visa / Mastercard Actually Cap Offline Contactless

Two different things get conflated in the popular narrative:

- The no-PIN convenience-tier cap (roughly £100 / ■20,000-equivalent) is a UX and risk-appetite choice, not a cryptographic limit. Above that threshold, EMV chip cards run full offline-authenticated transactions with no such ceiling.
- The deeper industry-wide reason offline payments generally stay capped is architectural: EMV, Visa, and Mastercard use a ledger/account model. The terminal doesn't hold the money — it trusts a remote balance it cannot check live. Counters and floor limits bound fraud exposure until the terminal reconnects. A consequence of THEIR architecture, not proof offline double-spend is unsolvable in general.

2. The Part That Really Is Unsolved (Be Precise)

In a ledger/account model, preventing double-spend with zero communication between verifying parties is provably impossible — a distributed-consensus problem, the same category that forces Bitcoin to need a blockchain instead of just signed messages. Nobody has solved this. We have not solved this. This is not what we're claiming.

3. Why Our Architecture Sidesteps the Problem Instead of Solving It

We do not use a ledger model. We use a stored-value / e-purse model — the same core idea behind Mondex and Hong Kong's Octopus card, real and proven for over 25 years. The spendable money physically exists as a cryptographically sealed number inside the customer's own PUF-secured chip. There is exactly one copy. When it's spent at Terminal A, it is gone from that chip — actually, verifiably decremented, right there. There is nothing left for Terminal B to double-spend, because Terminal B, tapping that same physical chip, sees the already-reduced balance.

4. Frame by Frame — What Actually Happens During a Single Tap

Frame	What Happens	What It Specifically Blocks
1 — Field Established	Customer taps at Terminal A. NFC field activates the secure element.	Sets up interaction; nothing decided yet.
2 — Fresh Challenge	Terminal issues a brand-new, unpredictable PUF challenge, never reused.	Defeats replay and pre-recorded relay setups.
3 — Unclonable Response	Chip returns its PUF response, proving it's this exact silicon, not a clone.	Blocks card cloning — no two PUF chips respond alike.
4 — Distance-Bounding	Timing round-trip check runs concurrently; rejects slow responses.	Blocks relay attacks from a distance.
5 — Reads Own State	Chip reads its own balance and monotonic counter, entirely on-chip.	No external balance to be inconsistent with.
6 — Internal Decrement	Chip computes new balance = old minus amount, counter +1, from its own state only.	Blocks forcing a fake/higher balance — chip won't sign a state it didn't derive itself.
7 — ECC P-256 Signing	Chip signs the new state; private key never leaves the secure element.	Proves authenticity + tamper-evidence of a valid, in-order decrement.
8 — Terminal Verifies	Terminal checks signature against known public key; completes if valid.	No contact with anyone needed — fully local verification.
9 — Queued for Settlement	Signed token held with 72-hour TTL, queued for the Solar Mesh Hub.	Bounds max time any transaction stays unsettled.
10 — Hub Sync Check	Hub checks counter sequence per chip ID on reconnect, flags gaps/duplicates.	Final safety net catching any residual-risk anomaly before settlement.

5. Answering the Meta AI Critique Directly

Meta AI's skepticism about “ECC P-256 signature + monotonic counter” is correct — for a specific framing. If you picture the counter and signature as something an independent verifier checks against a remote reference balance (the ledger model), counter + signature only stops replay of an old message to the same verifier. It does nothing to stop a fresh, validly-incrementing message being used at a different, never-before-seen verifier against the same remote balance. That really is still broken — Meta AI is right to flag it. This is the actual EMV-style problem.

The correct framing is different: the counter and signature are not a check performed by a verifier against a reference — they are generated by the chip AS the value store itself. Verification isn't “does this match a database,” it's “is this a validly-signed decrement of the one and only value that exists.” The counter proves order; the signature proves only this unclonable chip could have produced it. Combined, no external ledger is needed at all.

6. What's Genuinely New vs. 25-Year-Old Prior Art

How to read this table: prior art means an existing real-world system already uses the underlying idea; the honest claim is about which specific combination is new, not the base concept.

System	Architecture	Gap vs. ZerôPâñ Money
Mondex / Octopus (1990s-2000s)	Stored-value chip — same core idea we use	Static-key cards, cloneable with resources; no distance-bounding; no rural zero-infra settlement layer
EMV / Visa / Mastercard contactless	Ledger model, no-PIN convenience cap	Capped by design above a threshold; needs periodic online reconciliation

Moniepoint / OPay “offline-first”	Ledger model, queue-and-sync-later	Real fraud exposure window until queued transaction syncs
ZerôPâ Money	Stored-value + PUF (unclonable upgrade) + distance-bounding + Solar Mesh Hub settlement	Purpose-built combination for zero-connectivity rural NFC corridors — this combination is the new part

The accurate claim: we are not the first to use stored-value architecture to avoid double-spend. We ARE combining it with materially stronger anti-cloning (PUF vs. static keys) and anti-relay (distance-bounding) protection than any prior stored-value system had, purpose-built for a use case nobody has deployed this exact combination for. That is a real, precise, and still genuinely strong “world-first” claim.

7. What's NOT Fully Solved — Residual Risk, Stated Plainly

- PUF unclonability is an extremely strong, well-regarded hardware security property — not a mathematical proof. A sufficiently resourced attack (invasive probing, side-channel, fault injection) could theoretically still compromise a specific chip. This is exactly why independent lab attack-testing (\$60,000, already budgeted for Niger) matters — it tells us if our specific implementation actually holds up under real adversarial testing, not just on paper.
- If a customer's own legitimate device is compromised via firmware attack while genuinely offline, a skilled attacker might, in theory, force a stale state — bounded and later caught at hub sync via sequence-gap detection, not prevented in the instant it happens. A real, bounded exposure window, same risk category EMV manages via floor limits — we manage it via short TTL plus hub-side deduplication instead of a value cap.
- None of this replaces getting actual independent lab validation before making any public “solved” claim to press, investors at scale, or regulators. Internally we are confident in the architecture. Externally, “confident” and “independently verified” are different claims, and only the second one should ever be said publicly.

8. Recommendation

Frame this externally as: “architected specifically to remove the double-spend exposure window that forces industry-wide offline caps,” rather than “solved what nobody else has solved.” The first is technically airtight and still a powerful differentiator. The second invites a challenge we can't fully back until the lab report exists.

The \$60,000 independent lab validation line already built into the Niger State \$500,000 ask is the correct next real step — it's what turns “we believe we've cracked it” into “a third party confirms we've cracked it.” That's the document that should exist before any “world-beater” language goes anywhere near press, an acquirer, or a regulator.

9. Attack Timing — Two Attacks That Matter, In Milliseconds

Relay attack (attacker fakes physical presence over the internet):

- $t=0\text{ms}$: Attacker places a proxy device near the real terminal, a second proxy near the victim's device far away, relaying the NFC signal over internet/mobile data.
- $t=0-15\text{ms}$: Handshake proceeds normally — invisible at this stage, since relayed protocol messages look identical to genuine ones.
- $t=15-20\text{ms}$: Distance-bounding timing check runs. Genuine NFC coupling happens over $\sim 4\text{cm}$, completing well under a couple of milliseconds once protocol overhead is stripped out. Relaying over decent 4G adds 20-150ms; over satellite, 500-700ms — orders of magnitude outside the tight window a truly local tap should produce.
- Result: declined at roughly $t=20\text{ms}$, before the ECC signing step even runs — caught on timing physics alone, not cryptography.

Double-spend attempt (same offline value used at two terminals before any sync):

- $t=0\text{ms}$, Terminal A: payer taps. It is the payer's own device that spends the money — the payer's secure chip decrements its internal balance and commits a new signed state, with a hardware monotonic counter ticking forward. That commit makes the money gone from the payer's side, immediately.
- $t=X$, Terminal B: for double-spend to work, the attacker needs the payer's device to present its pre-transaction balance again — meaning roll back that chip's state, or clone it entirely.
- Rollback blocked: the hardware counter is monotonic — it can only count up, even under direct memory manipulation.
- Cloning blocked: PUF-derived key material comes from atomic-level manufacturing randomness in that one specific piece of silicon, never stored anywhere retrievable, not even by us as manufacturer.
- Result: the second tap produces an already-spent signature the receiving terminal rejects, or fails outright — caught locally, same $\sim 100-150\text{ms}$ window as any normal tap.

10. Solar Mesh Hub — Threat Model (The Honest Open Gap)

The hub is a relay and aggregator, not a vault — customer value lives inside each terminal's PUF-secured element, not the hub. Closer to a postal sorting office than a bank vault.

If a hub is physically stolen: the attacker gets a battery, charge controller, radio, and queued tokens — all still signed by the originating terminal's own key, which the hub never controlled. No spendable value, no private keys. Worst case is a delay, not fraud.

If a hub is tampered with in the field: this is the genuinely open question. Safety requires the hub to carry its own device identity so tampered or unrecognized hubs can be rejected at settlement — and depends on the originating terminal's signature (not the hub's) being checked end-to-end.

Open items, not yet formally closed — explicitly in scope for the \$60,000 independent lab validation before scaling past the Niger State pilot:

- Hub-level device attestation (certificate/keypair per hub, distinct from terminal keys).
- Key revocation process if a specific hub is confirmed compromised.
- Tampering resistance for the hub enclosure itself (physical hardening, not just cryptographic).

- Defined handling for tokens mid-relay through a hub at the exact moment of compromise.

This is stated plainly on purpose: it is a real, currently open gap, not a solved problem — exactly the kind of thing lab attack-testing exists to catch before wider deployment.

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